# Label Provenance and Metric Validity in Rule-Generated Corpora: Evidence from Eight Iterations of a Domain Fine-Tune

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***Abstract*—Rule-generated synthetic corpora are attractive where no labeled data exists: every label is correct by construction with respect to the encoded rules, and no language model fabricates domain content. We report on LoadBrief, a 50,000-example corpus pairing athlete-monitoring narratives with structured load-management briefs, and on eight successive fine-tunes of Llama 3 8B Instruct trained against successive versions of it. The final model reaches 0.960 exact risk-classification accuracy against a base model scoring 0.000. We show this measures less than it appears to, and that the instruments reporting it are as defective as the corpus they measure.**

**Both ground-truth labels are per-scenario constants rather than functions of the sampled physiological signals: across 40,000 records, each of 19 scenarios maps to exactly one risk level and one overreaching class with zero variance. A TF-IDF bag-of-words classifier recovers the risk label from the narrative at 0.950, within one point of the fine-tune, so the headline figure cannot be evidence of clinical inference. The construction admits defects that label-consistency validation cannot see: two disjoint families of self-contradictory records affect 16.1% of the corpus, and all three of the pipeline's independent consistency guards pass every one, each for a different structural reason.**

**Three evaluation instruments fail alongside the data. The composite reward's 0.701 ceiling is an artifact of three of five components being mis-specified against the corpus's own register conventions; an untuned model scoring 0.422 leaves it a usable range of 0.279. The overreaching metric cannot read 45% of ground-truth briefs, reframing a seven-revision "capability loss" as a generator change. And judge calibration against reference briefs shows the model tracks its training targets at $\rho = 0.83$, locating the reported clinical weakness in the data rather than the fine-tune. We release the artifacts unmodified with these defects documented, and propose four cheap checks — among them declared-target reachability auditing, and validating extraction metrics against reference outputs — that would each have caught a defect this project carried through multiple training cycles.**

***Index Terms*—synthetic data, instruction tuning, LoRA, label provenance, LLM-as-judge, evaluation methodology, sports science, athlete monitoring**

## I. Introduction

Wearable monitoring has made physiological data routinely available to athletes at every competitive level, but interpretation has not kept pace. Athlete monitoring aggregates external load, internal load, and subjective wellness into training decisions [14], [15], and translating an elevated acute-to-chronic workload ratio (ACWR) [1], a multi-day decline in heart-rate variability (HRV) [6], and a set of depressed wellness scores [9] into a defensible training decision requires expertise that most athletes and coaches cannot access. The underlying constructs are themselves contested [4], [5], which makes consistent, auditable application of the published criteria a prerequisite for any useful automation. General-purpose language models perform poorly at this task: they apply generic rather than sport-specific thresholds, reason inconsistently when monitoring signals conflict, and do not reliably hold an audience register through prompting alone.

No public dataset pairs athlete monitoring data with written interpretive briefs. We therefore constructed one, following the rule-based rather than model-generated route — the latter being standard for instruction tuning [16], [17] but prone to fabricating domain content and to recursive degradation [21]. LoadBrief is generated by a rule-based simulator that encodes four bodies of published sports science as quantitative thresholds, samples a 28-day monitoring time series per example, and renders a structured brief in three audience registers. Because no language model participates in generation, no clinical content is fabricated, and every label is correct by construction with respect to the encoded criteria. This is the standard argument for rule-generated corpora, and we made it ourselves in earlier releases of this work.

This paper is an account of why that argument is weaker than it looks. Our organizing claim is a single sentence: **rule-generated data is not self-validating, and neither are the metrics used to evaluate models trained on it.** "Correct by construction" guarantees only that a label agrees with the rule that produced it. It says nothing about whether that rule's output agrees with the data rendered alongside it, whether a declared generation parameter was reachable, or whether the metric scoring the result can recognize a correct answer when shown one. Each of those is a separate property requiring a separate check, and in this project every one of them went unchecked through multiple training cycles.

The findings that follow are five instances of that single failure, not five unrelated bugs. Two of them concern the corpus: labels that do not depend on the features rendered beside them, and declared parameter ranges the generator's own sampler could not reach. Three concern the instruments: a reward function mis-specified against its own reference data, a classification metric unable to read 45% of correct answers, and a judge whose absolute scores are uninterpretable without calibration against ground truth. The corpus defects are the more obvious kind; the instrument defects are the more dangerous, because they silently determined what the corpus defects looked like from the outside.

Over eight corpus revisions and eight corresponding supervised fine-tunes, we repeatedly improved headline metrics while carrying these forward. Each was detectable in seconds by a check we did not initially run.

Our contributions are:

1) **A label-provenance audit** of a released rule-generated corpus, showing that both ground-truth labels are per-scenario constants with zero within-scenario variance, and that the reported classification accuracy therefore measures scenario recovery rather than clinical inference (Section IV).
2) **Characterization of a defect class that defeats three independent consistency guards.** Two disjoint contradiction families affect 16.1% of the corpus. A severity-rank filter, a phrase-bank consistency checker, and a schema validator all pass them, for three unrelated structural reasons; we diagnose each blind spot. A stratified LLM-judge protocol localizes the damage precisely (Section V).
3) **A judge-calibration procedure that separates model error from data error.** Scoring the corpus's own ground-truth briefs with the identical judge, across 57 scenario-register cells, shows the model tracking its training targets at $\rho = 0.83$ and rules out a judge ceiling. The procedure also surfaces a register-level defect invisible to the main evaluation, and the judges independently identified the second contradiction family before it was measured (Section X).
4) **A decomposition of the composite reward function** showing that its empirical ceiling of 0.701 arises from three of five components being mis-specified against the corpus's own register conventions, not from task difficulty. This explains the otherwise-paradoxical result that reward fell across revisions while classification accuracy rose monotonically (Section XI).
5) **Self-extraction as a metric-validation check.** Scoring a corpus's ground-truth briefs with the evaluation metric that will be applied to model output reveals that the overreaching metric can read only 55% of correct answers, and none at all for one class. This reframes a seven-revision "capability loss" as an artifact of a generator change (Section XII).
6) **A reachability audit method** that compares each scenario's declared metric target against the distribution its own sampler produces, and which retrospectively catches seven defects that were each found separately and expensively (Section VI).

## II. Related Work

### A. Athlete monitoring and its decision rules

Athlete monitoring aggregates external load (distance, accelerometry), internal load (session RPE, heart-rate measures), and subjective wellness into decisions about how much an athlete should train [14], [15]. Foster's session-RPE method [11] made internal load cheap to collect and underpins most of the derived quantities used here, including training monotony and strain [10]. Heart-rate variability entered practice as a non-invasive index of autonomic recovery, with Plews et al. [6] establishing that individual baselines and multi-day rolling means, rather than single-morning readings, carry the usable signal [7]. The generator's combined-signal severity rule, in which converging adverse signals escalate risk beyond any single marker, follows the integrated-monitoring position of Buchheit and Laursen [12]. Subjective wellness instruments in the Hooper and Mackinnon tradition [9] remain competitive with, and by some accounts more sensitive than, objective measures [13]. Overreaching states are conventionally partitioned into functional overreaching, non-functional overreaching, and overtraining syndrome by the joint ECSS/ACSM consensus statement [8], which frames the distinction in terms of recovery timescale rather than any single marker.

The acute-to-chronic workload ratio [1], [2] is the most widely adopted of these constructs and also the most contested. Subsequent work has challenged it on mathematical coupling between numerator and denominator [3], on conceptual and methodological grounds [4], and on the analytic choices underlying the original injury-risk claims [5]. We take no position on that debate, but it bears directly on our contribution. A corpus whose labels are derived from ACWR thresholds inherits whatever validity those thresholds have, and we are careful throughout to claim only consistency with published criteria rather than clinical correctness. The defects we report are prior to that question: they concern whether the corpus is internally consistent with the rules it claims to encode, which is a necessary condition for the validity debate to matter at all.

To our knowledge no public dataset pairs athlete monitoring data with written interpretive briefs, which is why the corpus was constructed rather than collected.

### B. Synthetic data for instruction tuning

Instruction-tuned models are routinely trained on generated data. Self-Instruct bootstraps instruction-response pairs from a language model's own outputs [16]; Alpaca applied the recipe at scale [17]; Evol-Instruct increases instruction complexity through iterated rewriting [18]; Unnatural Instructions demonstrated that a modest seed set suffices [19]. The phi series argued that carefully generated "textbook-quality" data can outperform far larger web corpora [20].

The known hazard of this family is recursive degradation. Training successive generations on model-generated data narrows distributional tails and can collapse the learned distribution entirely [21], [22], though accumulating rather than replacing real data mitigates it [23]. A second hazard is fabrication: an LLM asked to generate clinical content will produce fluent content that is not grounded in any source.

Rule-based generation, as used here, is the natural response to both. No model participates, so nothing is fabricated and no recursive loop exists; every label is derived from an explicitly encoded criterion. Our contribution is to characterize what this substitution costs. Rule generation does not eliminate the data-quality problem so much as relocate it, from fabricated content to defects in the encoding: labels that do not depend on the features the generator renders alongside them, declared parameter ranges that the generator's own sampler cannot reach, and self-contradictory records that pass every consistency check the authors wrote. These failures are quiet in a way fabrication is not. Fabricated clinical content is visible on inspection; a label that is a generator constant is invisible in any individual record and appears only in aggregate.

### C. Dataset artifacts, shortcuts, and label provenance

Our central finding — that a headline accuracy figure measures recovery of a generator configuration rather than the capability it names — belongs to a well-established line of work on datasets that admit unintended solutions. Torralba and Efros showed that vision benchmarks carry enough dataset-specific signal for a classifier to identify the source corpus [24]. In NLP, hypothesis-only baselines solve a large fraction of natural language inference benchmarks without reading the premise [25], because crowdworker generation strategies leave lexical artifacts correlated with the label [26]; models exploit syntactic heuristics rather than learning entailment [27]. Geirhos et al. gather these under *shortcut learning* [28], and Lapuschkin et al. document Clever Hans predictors that achieve strong benchmark performance through incidental correlations [29]. Northcutt et al. show that even curated test sets contain pervasive label errors sufficient to reorder model rankings [30].

The check we propose in Section VI is, in software-testing terms, property-based testing applied to a data generator: a declared invariant is checked against the implementation's actual output over many random draws, in the manner of QuickCheck [31], and the comparison of a declared relation against realized behaviour is the same move metamorphic testing makes for programs without oracles [32], [33]. Data-validation systems for production pipelines apply related ideas to datasets rather than generators, checking schemas and distributional constraints [34], [35]. We claim no novelty for the technique. We claim that it is not applied to synthetic-corpus generators as a matter of course, and that in a generator running backward from labels its absence admits a defect class that label-consistency validation cannot see by construction.

Our case differs from these in provenance rather than in kind. The artifacts above arise from human annotation strategies or incidental collection correlations, and are discovered by adversarial probing. Ours arises from a generator specification — the shortcut is not an accident of collection but a property of the label definition — and is discoverable by inspecting the generator, at zero cost, if anyone thinks to ask what the label is a function of. This suggests a documentation obligation that current practice does not impose. Datasheets [36], data statements [37], and model cards [38] ask how data was collected, by whom, and for what purpose, but for synthetic corpora they do not ask the question that matters most here: for each label, is it computed from the features released alongside it, or assigned independently of them? We argue in Section IV that this should be stated explicitly, and in Section VI that declared generation parameters should be audited against what the generator actually produces.

### D. Parameter-efficient fine-tuning and reward optimization

All models here are LoRA adapters [39] over Llama 3 8B Instruct, trained on a single consumer device; quantized variants extend the same approach further [40]. The first release added a reinforcement learning stage using GRPO [41], which forgoes the value network of PPO [42] and, unlike preference-based methods such as DPO [43], suits tasks with a programmatic reward rather than human preference pairs [44]. Specialization of this kind degrades unrelated capability, a long-studied phenomenon [45] that persists in instruction-tuned LLMs [46]; the v1 release measured it on MedQA and MMLU.

Section XI concerns the reward function itself, and connects to work on reward misspecification. Optimizing a proxy that diverges from the intended objective is a canonical safety concern [47], formalized as reward hacking [48] and characterized empirically as a phase transition in which agents that score better on the proxy perform worse on the true objective [49]. Gao et al. quantify the analogous overoptimization in RLHF reward models [50]. Our result is a simpler and, we suspect, more common failure than the ones typically studied. The reward here is not gamed by the policy; it is mis-specified against its own reference data, such that ground-truth outputs forfeit three of five components for reasons traceable to the corpus's register conventions. The consequence is that the metric's empirical ceiling reflects a defect rather than task difficulty, and that a reported reinforcement-learning saturation result admits a mundane explanation. Reporting a reference ceiling — which we did — is a useful precaution that proves insufficient on its own; the ceiling must be decomposed before it can be interpreted.

### E. LLM-as-judge and stratified evaluation

Model-based evaluation is now standard where reference-based metrics are inadequate [51], [52], and its biases are documented: position and verbosity effects [53], self-enhancement bias [51], and a measurable preference for a judge's own generations [54]. We use a judge from a different model family for this reason, and report single-judge dependence as a limitation.

The methodological point we add is about disaggregation rather than judging. Aggregate scores conceal failures concentrated in subpopulations, which motivates behavioral testing [55], worst-group reporting, and the disaggregated protocols of large-scale evaluation suites [56], [57]. In our case the pooled judge score is uninformative — it moves little across eight revisions — while the same ratings stratified by scenario type localize clinical failure precisely to the strata containing the generator defect. More pointedly, the stratified judge found what three purpose-built deterministic checkers could not (Section V): the deterministic checks are precise within the defect taxonomy each encodes and silent outside it, whereas the judge is noisier but not bounded by a taxonomy fixed in advance. We read this as an argument for pairing them rather than treating model-based evaluation as a softer substitute for programmatic checks.

## III. Corpus Construction

### A. Generation pipeline

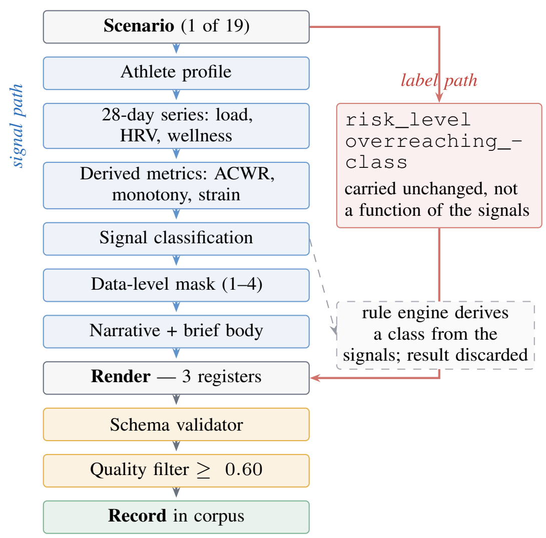


Fig. 1. The generation pipeline. The signal path (left) samples, simulates, derives, classifies and masks, then renders the narrative and the body of each brief. The label path (right) carries the scenario's declared `risk_level` and `overreaching_class` straight to the brief's risk header and classification section, bypassing all of it; the rule engine's own derived class is discarded (Section IV). The two paths meet only at render time, so nothing compares what the signals imply against what the label declares — the mechanism behind every defect in Section V.

Each example is produced by the sequence in Figure 1. A scenario is drawn from a weighted pool that repeats the tier-1, tier-2 and tier-3 scenario lists 8, 7 and 5 times respectively; because those lists hold 5, 5 and 9 scenarios, the resulting pool shares are 33.3%, 29.2% and 37.5%, not the 40/35/25 the multipliers suggest. An athlete profile is sampled, constrained by the scenario where the scenario declares constraints on athlete level or maturation status. A 28-day time series of session loads, HRV values, and five-dimension wellness scores is simulated to match the scenario's declared trajectory. Derived metrics are computed: ACWR by rolling 7:28-day means, training monotony, and strain. HRV trend, wellness composite, and ACWR zone are classified against the configured thresholds. A data completeness level (1–4) is applied, masking signals an athlete at that monitoring tier would not possess. A narrative is rendered from the surviving signals, and three audience-conditioned briefs are generated. Validation and quality filtering reject examples below threshold.

Candidate examples are oversampled by a factor of 1.15 and passed through two independent gates before entering the corpus. A schema validator checks required metadata, narrative length and content, per-register brief structure, label validity, and one cross-field consistency condition. A quality filter then scores each example on five weighted dimensions — recommendation specificity 0.25, narrative informativeness 0.20, audience differentiation 0.20, clinical coverage 0.20, and language naturalness 0.15 — and rejects anything below a composite threshold of 0.60. The released corpus has a mean quality score of 0.8585, well clear of the threshold, which indicates the filter is removing a tail rather than shaping the bulk of the distribution. Both gates are analyzed as consistency guards in Section V; neither detects the defects reported there.

Generation runs *backward from the label*: the scenario's classification is fixed first and the time series is sampled to match. Section IV examines what this implies, and Section IV-B tests empirically how easily that configuration can be recovered. A complete example record appears in Section III-B.

### B. A worked example

Figure 2 shows a complete record from the released corpus. It is a `fixture_congestion` scenario at data completeness level 2, rendered in the coach register. We reproduce it in full because every defect analyzed later in this paper is visible as a relationship between parts of a record like this one, and those relationships are difficult to follow in the abstract.

Two features are worth noting now. The narrative reports every derived metric the generator computed, so a model has access to the same quantities the label was nominally derived from. And the recommendation — a 40–50% volume cut over 5–7 days — is prescribed for a presentation the brief itself describes as having optimal ACWR and good wellness across all dimensions. That tension is mild here. Section V shows what it becomes when the sampled signals and the declared label diverge further.

### C. Corpus composition

Table I summarizes the released v8 training split (40,000 records; validation and test splits of 5,000 each under an 80/10/10 partition with seed 42). The corpus spans 66 sports in 10 categories, 19 scenario types, and 11 training phases. Mean composite quality score is 0.8585 against a rejection threshold of 0.60. Realized tier shares are 34.5%, 30.2% and 35.3%, close to the pool shares above; the gap is the quality filter, which rejects unevenly across tiers.

**input_narrative**

Monitoring snapshot: Recreational Swimming, July 24 – August 21, 2026. Athlete profile: amateur, training age 5 years, phase: pre-season early. The acute:chronic ratio (1.226) is in the sweet spot, indicating appropriate progressive loading. Acute load (past 7 days): 128 AU. Chronic load (28-day average): 104 AU. Training monotony (0.96) is within normal range. A 6.4 ms drop in heart rate variability below baseline suggests early fatigue accumulation. Wellness is good across all dimensions — no subjective readiness concerns. Average nightly sleep of 7.9 hours. Context: competitive schedule has required 2 matches per week during this monitoring period.

**output_coach**

`RISK LEVEL:` `MODERATE`. Some monitoring signals warrant attention.
`TRAINING LOAD:` ACWR of 1.226 is within the optimal training range. Acute load: 128 AU | Chronic load: 104 AU.
`PHYSIOLOGICAL READINESS (HRV):` HRV moderately suppressed: 33.2 ms (6.4 ms below 48.2 ms baseline).
`SUBJECTIVE WELLNESS:` All monitored wellness dimensions within normal range. Composite status: good.
`SIGNAL INTEGRATION:` Monitoring signals are largely positive with one area of concern: HRV mildly suppressed. Other signals remain within acceptable ranges.
`CLINICAL CLASSIFICATION:` Functional overreaching — short recovery period (3–7 days) will restore performance.
`RECOMMENDATIONS:` Significant load reduction required: cut volume by 40–50%, eliminate all high-intensity work for 5–7 days.
`MONITORING PRIORITIES:` Increase monitoring frequency: daily HRV + pre-session wellness questionnaire.
`ESCALATION TRIGGERS:` Escalate to physician review if HRV suppression persists beyond 7 days; performance declines despite load reduction; athlete reports inability to complete prescribed sessions.

**labels** `risk_level: moderate_to_high` `overreaching_-classification: functional_overreaching` `acwr_value: 1.226`

Fig. 2. A complete record from the released v8 corpus, coach register, abridged only in the athlete profile line. Note that `RISK LEVEL` renders the scenario constant `moderate_to_high` as `MODERATE` (Section IV), and that the recommendation prescribes a 40–50% volume cut for a presentation the same brief describes as being in the optimal training range with good wellness.

TABLE I
V8 TRAINING SPLIT COMPOSITION ($n = 40{,}000$)

| **Complexity tier** | | **Data level** | |
|---|---|---|---|
| Tier 1 (clear-cut) | 13,805 | Level 1 (minimal) | 6,256 |
| Tier 2 (moderate) | 12,063 | Level 2 (collegiate) | 15,972 |
| Tier 3 (conflicting) | 14,132 | Level 3 (professional) | 13,895 |
| | | Level 4 (research) | 3,877 |
| **Risk label** | | **Overreaching label** | |
| moderate | 15,829 | normal adaptation | 23,008 |
| moderate–high | 6,617 | functional overr. | 13,570 |
| low | 5,489 | non-funct. overr. | 1,716 |
| high | 4,540 | overtraining synd. | 1,706 |
| low (detraining) | 2,836 | | |
| low (if managed) | 2,656 | | |
| critical | 1,706 | | |
| variable | 327 | | |

### D. Audience registers

Each monitoring narrative is rendered into three briefs targeting different readers. This is a design goal of the system, not a formatting convenience: the premise is that an athlete, a coach, and a sports scientist need the same clinical content at different levels of technical vocabulary, and that a practitioner-facing tool should produce all three from one set of measurements.

TABLE II
PER-REGISTER GENERATION TARGETS. TECHNICAL DENSITY IS THE FRACTION OF TOKENS DRAWN FROM A CLINICAL VOCABULARY; FK IS FLESCH–KINCAID GRADE LEVEL.

| Register | FK target | Tech. density | Words |
|---|---|---|---|
| athlete | 6–9 | $\leq 0.05$ | 150–250 |
| coach | 9–12 | $\leq 0.12$ | 250–400 |
| sports scientist | 12–16 | $\leq 0.25$ | 350–600 |

The generator encodes explicit, quantitative targets for each register (Table II), and an audience adaptation layer enforces them by substituting vocabulary, trimming to length, and adjusting tone. Athlete briefs receive lay substitutions — "recovery score" for HRV, "training spike ratio" for ACWR, "overtraining warning" for non-functional overreaching — and open `YOUR STATUS:`. Coach briefs retain clinical terms and open `RISK LEVEL:`. Sports-scientist briefs add literature citations and open `OVERALL RISK CLASSIFICATION:`. The header divergence is verified across all 40,000 records with no exceptions.

The adaptation does what it was built to do: the registers are lexically distinct, and the athlete register avoids the terms it is configured to avoid. Two later findings concern what that costs. Section XI shows the lay substitution forfeits 86% of one reward component, because the reward matches on the clinical strings the adaptation removes. Section X shows the athlete register's ground-truth briefs score 0.88 lower on judged clinical accuracy than the coach register's, and are lowest of the three in 16 of 19 scenarios. Neither result is a failure to meet the targets in Table II. Both indicate that meeting them, as implemented, removes clinical content rather than restating it — a distinction the design assumed away and neither gate in the pipeline was positioned to test.

Formatting for training expands each record into three examples, one per register, tripling the effective corpus.

## IV. LABEL PROVENANCE

### A. Both labels are scenario constants

The generator writes the ground-truth risk level and overreaching classification directly from the scenario definition: `scenario.risk_level` and `scenario.overreaching_class`. Neither is a function of the sampled physiological signals.

The rule engine does compute an independent classification from the signals, comparing ACWR zone, consecutive HRV suppression days, and wellness composite against the Meeusen criteria. It then discards the result and returns the scenario constant. A source comment states that mismatches are "logged for quality filtering"; no such logging exists in the released pipeline. A later revision added a rejection filter for undeclared label divergence, which addresses generation-time divergence but not the structural fact that the label never derived from the signals in the first place.

We verified this empirically. Table III cross-tabulates scenario type against both labels and against the rendered risk header over all 40,000 records. Every scenario maps to exactly one risk label, one overreaching label, and one rendered header, with zero variance.

### B. What classification accuracy measures

It follows that recovering the risk label is equivalent to recovering the scenario type, up to the many-to-one collapse of 19 scenarios onto 8 risk labels. Whether that recovery is difficult is an empirical question, and we test it rather than assume it.

We fit a TF-IDF bag-of-words model (unigrams and bigrams, minimum document frequency 3) with multinomial logistic regression to predict each label directly from `input_narrative`, on an 80/20 stratified split of the released corpus. The classifier sees only the narrative — the same text the fine-tuned model receives — and never sees a brief. It has no capacity for clinical reasoning of any kind: it is a linear function of word counts.

Three things follow from Table IV.

**Scenario identity is trivially recoverable.** A linear model identifies which of 19 generating scenarios produced a narrative 90.5% of the time against a 7.1% majority baseline. Because both labels are scenario constants (Section IV), recovering the scenario is sufficient to recover the labels. The premise of this section is therefore established rather than assumed: the narrative carries the generator's configuration in a form a bag of words can read.

**The headline accuracy is matched by a linear model.** Logistic regression on word counts reaches 0.950 exact and 0.991 within-one-class risk accuracy, against the fine-tune's 0.960 and 0.994. We do not claim the two systems perform the same task — the fine-tune generates a complete register-conditioned brief and the classifier emits a class label, and only the former is the product. But on the specific quantity reported as this system's headline result, a model with no capacity for inference of any kind is within one point. Whatever the 0.960 demonstrates, it is not evidence of clinical reasoning, because clinical reasoning is not required to produce it.

**The overreaching label is present but unmeasured.** The linear model recovers overreaching class at 0.971 while the fine-tune is reported at 0.504. The information is in the narrative and is nearly perfectly extractable from it; the reported figure reflects the generate-then-extract measurement pipeline rather than the availability of the signal. Section XII diagnoses that pipeline directly and arrives at the same conclusion from the opposite direction.

Two caveats. The split is of the training corpus rather than the formatted test split, since the classifier consumes raw narratives; leakage between them is not possible because the classifier never sees a brief. And discriminative classification is an easier problem than conditioned generation, so Table IV bounds what the accuracy metric establishes, not what the model is capable of. That bound is the point: a metric a linear model saturates cannot distinguish a system that reasons from one that pattern-matches, and should not be presented as though it does.

The distinction is not merely interpretive. Because the label is a constant, it cannot respond to the sampled signals *even when they contradict it* — which is the mechanism behind the defects in Section V.

### C. An apparent severity ceiling that is not one

All 1,706 records labeled `CRITICAL` fall below ACWR 0.8, and no record above ACWR 0.8 is ever labeled `CRITICAL`. Read naively this suggests a severity ceiling on high-load presentations. It is not. The `CRITICAL` records are exactly the 1,706 `overtraining_syndrome` records — the identity is exact — and the OTS scenario declares an ACWR target of 0.45–0.65, modeling the depleted phase in which chronic overload has already suppressed training capacity. The ACWR distribution of the `CRITICAL` class is a property of that single scenario, not of a severity rule.

The consequence for a downstream model is nonetheless real: because `CRITICAL` is reachable only through one low-ACWR scenario, a model trained on this corpus will not assign `CRITICAL` to a high-load presentation under any physiological condition. We report the mechanism rather than the naive reading because the two imply different fixes.

## V. Defects Invisible to Label-Consistency Validation

### A. Adverse prose under a benign header

In 2,412 records (6.0% of the training split), the brief reports critically suppressed HRV in its physiological-readiness section while declaring `RISK LEVEL: LOW`. A representative coach brief reports HRV at 58.6 ms, 11.9 ms below individual baseline for seven consecutive mornings, states in its signal-integration section that objective HRV indicates physiological fatigue and that HRV should be trusted over the subjective report, and then classifies the athlete under normal adaptation at low risk.

The distribution is given in the final column of Table III. By ACWR band, 1,865 fall below 0.8 and 547 between 0.8 and 1.3; none occur above 1.3. It concentrates in `undertraining` (1,207) and `post_competition` (1,038) but is not confined to either.

The mechanism is the one established in Section IV. The narrative and the physiological sections are rendered from the sampled signals. The risk header is a scenario constant. When the sampler produces adverse physiology inside a scenario whose declared label is benign, the two halves of the brief disagree, and nothing in the pipeline compares them.

### B. A second contradiction axis

The pattern above is not confined to the risk header. The same mechanism — signal-driven prose meeting a constant label — produces a structurally identical defect on the overreaching axis, in a disjoint set of scenarios.

The `SIGNAL INTEGRATION` section is synthesized from the sampled signals, while `CLINICAL CLASSIFICATION` renders the scenario constant. When the sampler produces a

TABLE III

SCENARIO-TO-LABEL MAPPING IS DETERMINISTIC. EVERY SCENARIO YIELDS ONE RISK LABEL, ONE RENDERED BRIEF HEADER, AND ONE OVERREACHING CLASS ACROSS ALL 40,000 RECORDS, WITH ZERO WITHIN-SCENARIO VARIANCE. THE FINAL COLUMN COUNTS RECORDS WHOSE BRIEF REPORTS CRITICALLY SUPPRESSED HRV UNDER A `LOW` HEADER.

| Scenario | $n$ | Risk label | Header | Overreaching class | Contra. |
|---|---|---|---|---|---|
| undertraining | 2836 | low_with_detraining_flag | LOW | normal_adaptation | 1207 |
| acwr_spike | 2824 | high | HIGH | functional_overreaching | 0 |
| normal_progressive | 2767 | low | LOW | normal_adaptation | 165 |
| taper | 2722 | low | LOW | normal_adaptation | 2 |
| post_competition | 2656 | low_if_managed | LOW | normal_adaptation | 1038 |
| fixture_congestion | 2506 | moderate_to_high | MODERATE | functional_overreaching | 0 |
| preseason_intensification | 2425 | moderate_to_high | MODERATE | functional_overreaching | 0 |
| monotony_problem | 2396 | moderate | MODERATE | functional_overreaching | 0 |
| illness_return | 2394 | moderate | MODERATE | normal_adaptation | 0 |
| altitude_camp | 2342 | moderate | MODERATE | normal_adaptation | 0 |
| youth_growth_spurt | 1767 | moderate | MODERATE | normal_adaptation | 0 |
| wellness_crash_normal_load | 1749 | moderate | MODERATE | normal_adaptation | 0 |
| high_acwr_stable_physiology | 1733 | moderate | MODERATE | functional_overreaching | 0 |
| heat_acclimatization | 1731 | moderate | MODERATE | normal_adaptation | 0 |
| travel_jet_lag | 1717 | moderate | MODERATE | normal_adaptation | 0 |
| early_overreaching | 1716 | high | HIGH | non_functional_overreaching | 0 |
| overtraining_syndrome | 1706 | critical | CRITICAL | overtraining_syndrome | 0 |
| double_session_accumulation | 1686 | moderate_to_high | MODERATE | functional_overreaching | 0 |
| recreational_minimal_data | 327 | variable | VARIABLE | normal_adaptation | 0 |

TABLE IV

LABELS PREDICTED FROM THE MONITORING NARRATIVE BY TF-IDF PLUS LOGISTIC REGRESSION, 80/20 STRATIFIED SPLIT OF THE V8 CORPUS. THE LINEAR MODEL MATCHES THE 8B FINE-TUNE ON RISK CLASSIFICATION AND VASTLY EXCEEDS IT ON OVERREACHING.

| Target | Linear | Majority | Fine-tune |
|---|---|---|---|
| scenario type (19 classes) | 0.905 | 0.071 | — |
| risk level, exact (5) | 0.950 | 0.561 | 0.960 |
| risk level, within one class | 0.991 | — | 0.994 |
| overreaching class (4) | 0.971 | 0.575 | 0.504 |

presentation more severe than the scenario declares, adjacent paragraphs of the same brief assert different classifications. A representative `acwr_spike` record, whose label is `functional_overreaching`, reads:

> **SIGNAL INTEGRATION:** All primary monitoring signals are converging to indicate significant fatigue accumulation. Training load, physiological readiness, and subjective wellness are simultaneously flagging — this pattern is consistent with non-functional overreaching and requires immediate attention.
>
> **CLINICAL CLASSIFICATION:** Functional overreaching — short recovery period (3–7 days) will restore performance.

Measuring this across the corpus, 3,810 records (9.5%) name one overreaching class in the integration narrative and a different one in the classification section. A further 211 `high_acwr_stable_physiology` records (12.2% of that scenario) assert converging signals consistent with non-functional overreaching while simultaneously stating the load is well tolerated.

Figure 3 combines both axes. The two families are *disjoint*: no record carries both, and they affect non-overlapping scenarios. Together they affect 6,433 records, 16.1% of the corpus.

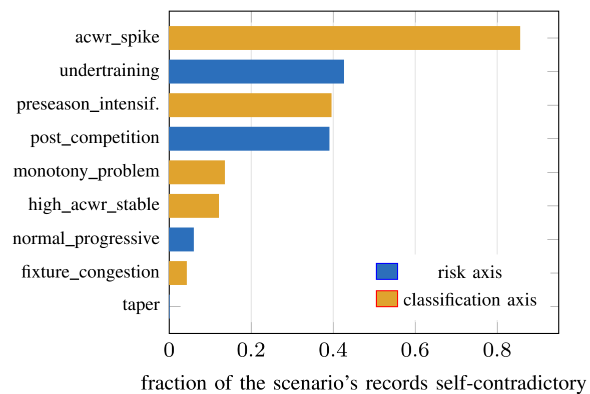


Fig. 3. The two contradiction families by scenario. "Risk axis" counts briefs reporting critically suppressed HRV under a `LOW` header; "classification axis" counts briefs whose signal-integration narrative names a different overreaching class than the classification section. The families are disjoint: no scenario carries both. Together they affect 6,433 records, 16.1% of the corpus. The ten scenarios not shown are clean.

That the two families are disjoint and hit different scenarios is itself evidence for the common cause. Neither is a localized coding error; each is the consequence of a label that cannot respond to the signals rendered beside it, surfacing wherever the sampler happens to disagree with the declaration.

### C. Three guards, three blind spots

The pipeline implements three independent consistency mechanisms, each written in response to an observed defect. All three accept every one of the 2,412 records (Table V). The reasons do not overlap.

TABLE V
GUARD OUTCOMES OVER THE RELEASED V8 CORPUS. NO GUARD FLAGS ANY OF THE 2,412 CONTRADICTION RECORDS.

| Guard | Flagged | Population |
|---|---|---|
| Label-agreement filter (gap $\geq 2$) | 0 | 40,000 records |
| Consistency checker (phrase banks) | 154 | 120,000 briefs |
| Schema validator (cross-field) | 0 | 120,000 briefs |
| *Consistency-checker flags by scenario* | | |
| post_competition | 106 | 7,968 |
| normal_progressive | 31 | 8,301 |
| undertraining | 17 | 8,508 |
| all other scenarios | 0 | — |

**The label-agreement filter** rejects samples whose rule-derived overreaching class diverges from the declared label by two or more severity ranks, unless the scenario declares the conflict intentional. Run over the released corpus it rejects 0 of 40,000 records. This is partly expected — v8 was generated with the filter enabled, so survivors pass by construction — but the structural reason matters more. Its severity ranking places `undertraining` at rank 0 alongside `normal_adaptation`, a deliberate choice documented in the source: `rule_class` returns `undertraining` whenever the ACWR zone is undertraining, which is expected during taper, post-competition, and critically in the depleted low-load presentation of overtraining syndrome, so ranking it as severe would reject every legitimate OTS sample. The consequence is that for an `undertraining` record the declared label is `normal_adaptation` (rank 0) and the rule-derived class is `undertraining` (rank 0), giving a gap of 0 regardless of how suppressed the sampled HRV is. The filter also operates only on the overreaching axis; the risk header is outside its scope entirely.

**The consistency checker** is a phrase-bank matcher built specifically to catch briefs whose body contradicts their classification. It is designed not to reject legitimate override scenarios, in which suppressed signals correctly coexist with a benign label because the brief *explains* the override, and implements this via an exculpatory clause: an adverse per-signal report does not count as a contradiction when the brief also contains conflict-acknowledgment language ("notable discrepancy", "trust HRV over subjective", "subclinical fatigue"). The defective briefs contain exactly that language. They acknowledge the discrepancy in prose and declare `LOW` anyway. The escape hatch that makes the checker safe for override scenarios is what makes it blind here. Over the released corpus it flags 154 of 120,000 register-expanded briefs (0.1%), all of them all-clear claims.

**The schema validator** includes a cross-field consistency check, but it is one-directional by construction: it verifies that a `high` or `critical` label is reflected somewhere in the brief text, and applies no check in the opposite direction. A benign label over adverse prose is unreachable for it.

Each guard is sound within its own frame. The filter reasons about label severity and not about prose; the checker reasons about prose and treats acknowledgment as sufficient; the validator reasons about under-statement of severe labels and not about over-statement of benign ones. The defect lives in the space none of them covers: a benign *constant* label, over prose that is adverse, correctly described, and honestly acknowledged — and still wrong at the header.

We draw a general lesson rather than a criticism of these particular implementations. Each guard encodes a defect taxonomy fixed when it was written. Adding guards of this kind increases coverage within the union of those taxonomies and provides none outside it, while producing a misleading impression of thoroughness: three passing checks read as strong evidence of consistency when they are three narrow ones.

### D. A guard exempted rather than satisfied

The three guards above share a structural weakness beyond their bounded taxonomies: two of them take their scope from a field in the data they inspect. The label-agreement filter exempts unconditionally any scenario whose `conflicting_signals` flag is set, on the reasoning that a declared override is intentional and the rationale layer explains it. That flag is written by the generator and exported per record, which makes it auditable — and makes visible an episode that would otherwise be untraceable.

Run against the surviving pre-filter v1 corpus, the filter rejects 1,651 of 40,000 records (4.1%), falling entirely within a single scenario, `early_overreaching`, at a rate of 100%. Every other scenario is untouched. The all-or-nothing pattern shows the filter keys on a scenario-level property, and identifies it precisely: `early_overreaching` declares `non_functional_overreaching` (severity rank 2) while its ACWR target of 0.9–1.2 lies in the sweet-spot zone, from which the rule-derived class is `normal_adaptation` (rank 0). Every sample clears the two-rank threshold.

The scenario does not trigger the filter in v8, and it was not repaired. The `conflicting_signals` flag is `False` for all 1,651 `early_overreaching` records in v1 and `True` for all 1,716 in v8. Recomputing the rule-derived class over the released corpus, all 1,716 records still carry a severity-rank gap of exactly 2 — 1,715 resolve to `normal_adaptation` against a declared `non_functional_overreaching` label — and every one would be rejected today had the flag not been changed. The same flag routes the scenario into the rationale layer, which is what makes the consistency checker exempt it as well. A single boolean silenced two independent guards while the underlying label-signal divergence remained exactly as it was.

We report this without imputing intent. Declaring a genuine signal conflict is a legitimate use of that flag, and `early_overreaching` — sustained elevated load producing accumulated fatigue at a normal acute-to-chronic ratio — is a plausible candidate for one. The structural point does not depend on the intent: when a guard's scope is controlled by a field in the data it inspects, satisfying the guard and repairing the condition it was written to detect are indistinguishable from the pipeline's perspective, and no artifact records which occurred. We could reconstruct this episode only because the flag happens to be exported per record. Had it lived solely in the generator configuration, which was not under version control, the v8 corpus would show a clean guard and no trace of how it got that way.

### *E. Why the judge found what the checkers could not*

The stratified LLM-judge results in Section IX localize clinical weakness to precisely the strata these defects occupy, and Section X shows the judges identified the second contradiction family from free-text comments before we measured it. This is the practical argument for pairing deterministic and model-based evaluation, and it generalizes past this corpus. A deterministic checker encodes a defect taxonomy fixed when it was written: within that taxonomy it is exact, cheap, and exhaustive, and outside it it is silent in a way that is indistinguishable from a clean result. A model-based judge has no fixed taxonomy. It is noisier, its absolute scores require calibration (Section X), and it cannot be trusted to be exhaustive — but it is not blind by construction to defect classes nobody anticipated. The two failure modes are complementary, and neither method's output should be read as evidence about the other's blind spots.

## VI. Declared-Target Reachability

A separate failure mode affected seven scenarios across earlier corpus revisions. Because generation runs backward from the label, a scenario declares a target metric range and a load pattern is supposed to produce it. Nothing checked that the pattern *could*.

Figure 4 reports the audit over every surviving corpus revision. This defect is invisible to label-consistency validation by construction: the label is consistent with whatever was sampled. It is the *declaration* that is unreachable, and nothing compared the two. Each of the seven was found separately — by a label-agreement filter, by judge comments, by a metric audit, by residual analysis — and six of the seven surfaced only after a model had already trained on the affected data.

We implemented a reachability audit that, for each scenario, compares the declared target range against the distribution present in a generated corpus and reports the fraction of samples landing inside it, classifying each as reachable, poorly centered, or unreachable. It exits nonzero when any target is unreachable, so it can gate a build. Running it retrospectively over all eight revisions produces Figure 4: nine targets were defective at v1 and eight were repaired by v8, in four distinct repair events at v4, v5, v6, and v7. Every one was detectable by this check in seconds, and each was instead found separately and expensively.

### *A. What reachability repair did and did not buy*

The lower panel of Figure 4 places the repair trajectory beside the evaluation metrics, and the contrast is the paper's clearest evidence for the label-provenance thesis.

Risk accuracy tracks reachability repair closely, rising from 0.792 to 0.960 as defective targets fall from nine to one, with the largest single jump (0.856 to 0.934) at v4→v5, exactly

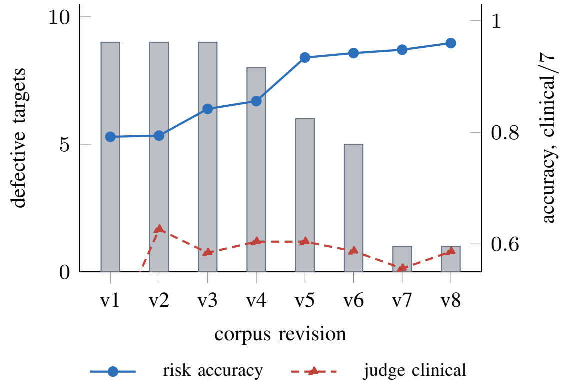


Fig. 4. Reachability repair against what it bought. Grey bars (left axis): the number of scenario targets whose declared range the sampler could not reach, falling from nine to one across four repair events. Solid line: exact risk accuracy, which tracks the repairs closely. Dashed line: judged clinical accuracy, which does not move. Making scenarios produce their declared ranges made them easier to recognize; it did nothing to make the constant labels respond to the signals, which is what clinical quality depends on.

where `acwr_spike` and `undertraining` were repaired. This is the expected consequence of Section IV: risk accuracy measures recovery of the generating scenario from its narrative, and making a scenario actually produce its declared metric range makes it far more distinguishable in the narrative text.

Judged clinical accuracy does not move. It peaks at v2 (4.38), sits at 4.10 in v8, and shows no trend across the entire repair sequence. Overreaching accuracy likewise fluctuates without direction and never reaches its majority-class baseline (Section XII).

Repairing eight of nine unreachable targets therefore bought 17 points of risk accuracy and no measurable clinical quality. That is precisely what the label-provenance account predicts. Reachability repair makes the sampled signals consistent with the scenario's declared *metric range*; it does nothing to make the ground-truth *label* respond to those signals, because the label never depended on them. The defect in Section V survives every one of these repairs untouched.

We recommend the audit as routine practice for any corpus generated backward from labels, with the caveat that passing it is necessary and not sufficient: v8 passes on eight of nine targets while carrying 2,412 self-contradictory records.

The audit as implemented covers only numeric declared targets (ACWR and monotony). Scenarios whose labels rest on HRV-day counts or wellness composite thresholds are not covered, which is a limitation: the defect in Section V involves HRV and would not be caught.

The same class of defect reaches beyond scenario targets. The generator's scenario pool is built by repeating the three tier lists 8, 7 and 5 times, which yields 40/35/25 only if the lists are equal in length. They are not — they hold 5, 5 and 9 scenarios — so the realized pool is 33.3/29.2/37.5, and the source comment recording the intended sampling distribution has been wrong since the first revision. Nothing in the pipeline compares a declared sampling weight against the corpus it produces, so this went unnoticed through all eight revisions and is reported here for the first time. It is minor in effect and exact in kind: a declared generation parameter that the generator does not produce, invisible to every check the project ran.

## VII. Training Methodology

### A. Supervised fine-tuning

All eight model revisions use identical supervised fine-tuning configuration on Llama 3 8B Instruct with LoRA. Rank 32, $\alpha$ 64, dropout 0.05, applied to all attention and MLP projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`). Learning rate $2 \times 10^{-4}$, cosine schedule, 3% warmup, weight decay 0.01, AdamW. Per-device batch size 2 with gradient accumulation 16 (effective batch 32), maximum sequence length 512, gradient checkpointing enabled, one epoch. Evaluation and checkpointing every 100 steps with best-checkpoint selection by validation loss.

Training used a 20,000-example subset of the register-expanded training split and a 2,000-example validation subset, capped to fit the memory budget of the training device. Models were trained in float16 on Apple Silicon (M4, 48 GB unified memory) via the MPS backend.

We emphasize that this configuration was held fixed across all eight revisions. Every run used LoRA rank 32, learning rate $2 \times 10^{-4}$, one epoch, 20,000 examples, and maximum sequence length 512; the recorded training configuration is byte-identical across the eight checkpoints apart from wall-clock. **The only variable across the revision sequence is the corpus.** The metric trajectories in Section IX — risk accuracy rising from 0.792 to 0.960 while judged clinical accuracy stays flat — are therefore attributable to corpus changes and cannot be explained by drift in training configuration.

Wall-clock per revision ranged from 28.3 to 36.1 hours (mean 31.4), totalling 251.4 hours of supervised fine-tuning across the eight revisions. These figures cover successful final runs only. They exclude resumed runs, the 500-completion generation pass required to evaluate each revision, all LLM-judge calls, and the reinforcement-learning configurations of the v1 release, whose checkpoints were not retained. Total project compute is therefore substantially higher than the figure reported here.

### B. Reinforcement learning (v1 only)

The first release added a GRPO stage optimizing the composite reward of Section XI: learning rate $5 \times 10^{-6}$, KL coefficient $\beta = 0.1$, 4 generations per prompt, temperature 0.8, top-$p$ 0.95, maximum completion length 400 tokens, 2 epochs, LoRA configuration inherited from SFT. GRPO was selected over PPO for the absence of a value network (roughly halving memory) and over preference methods for the availability of a programmatic reward.

None of the reinforcement-learning checkpoints were retained, so the v1 GRPO results are reported here as previously published rather than re-verified; the hyperparameters above are recovered from the training script. The v1 release reported that GRPO added no measurable benefit over SFT alone across eight configurations spanning multiple seeds and learning rates, plus two reward ablations, with an LLM judge agreeing within 0.04 on quality dimensions. Two configurations sharing a seed but differing hundredfold in learning rate reportedly produced identical training curves. Revisions v2–v8 are consequently SFT-only, and the released v8 model contains no RL stage.

Section XI supplies a mechanism for that saturation that the original report did not identify: the reward function's ceiling is set by components the ground-truth briefs themselves cannot earn, so a policy trained to maximize it has little headroom to exploit.

## VIII. Evaluation Protocol

Three instruments are used: a composite reward, a set of exact-match classification metrics, and a stratified LLM-as-judge protocol. Each rests on assumptions about the corpus, and Sections XI–XII show that all three assumptions fail in ways that were invisible until tested. We state them explicitly here so the later failures read as violated preconditions rather than surprises.

### A. Generation

For each model, 500 completions are generated greedily (`do_sample=False`, maximum 400 new tokens) from the held-out test split of that model's own corpus revision. Greedy decoding is used so that repeated evaluation of a checkpoint is deterministic; one sampled variant (`sft_v2_sampled`) is retained in the results below to show the decoding effect, which is substantial.

### B. Classification metrics

Risk level and overreaching classification are extracted from the generated text and compared against the record's labels. Risk extraction is register-aware, matching `RISK LEVEL`, `YOUR STATUS`, or `OVERALL RISK CLASSIFICATION` headers, and returns *unknown* rather than guessing when no header is present. Boundary labels are scored against accept-sets: `moderate_to_high` accepts either `moderate` or `high`; `low_with_detraining_flag` and `low_if_managed` accept `low`. We also report within-one-class accuracy, a count of severe errors (predictions two or more classes distant), and a count of unparseable headers.

*Assumption.* Extraction-based classification metrics assume the corpus phrases its classifications in a form the extractor can recognize — that a correct brief is a readable brief. Section XII shows this holds for risk level and fails for overreaching, where the extractor cannot read 45% of ground-truth briefs and 100% of one class.

### C. Composite reward

The reward assigns credit for a structured risk declaration, recommendation specificity, clinical overreaching terminology, an escalation trigger, and output length in an expected band; the full definition is given in Section XI. The same function served as the GRPO reward in v1 and as an evaluation metric throughout.

*Assumption.* A reward used as a quality metric assumes that outputs of the quality it is meant to reward can earn its available credit. Because the reward is not normalized to the task, we score the ground-truth briefs themselves and report that value as a reference ceiling beside every model result, so that scores are read against what the corpus can achieve rather than against 1.0. Section XI shows this precaution was necessary and, on its own, insufficient: a ceiling below 1.0 signals that the metric is unnormalized but does not distinguish task difficulty from metric defect, and here the gap is almost entirely defect.

### *D. LLM-as-judge*

Briefs are rated by Gemini 2.5 Flash on three 1–7 integer dimensions — clinical accuracy, actionability, and clarity — with a one-sentence justification of the lowest score. A model family different from the system under test is used to reduce self-preference bias. Sampling is stratified by scenario type up to a fixed number of briefs per scenario, so that rare scenarios receive enough ratings to support stratified analysis; all reported judge runs rate 438 briefs. The pooled "overall" figure is the mean of the three dimensions and is not an independent measurement.

*Assumption.* A judge score is interpretable as model quality only if the judge would score a correct answer highly. Nothing in the protocol establishes that. Section X tests it directly by scoring the corpus's own ground-truth briefs with the identical judge, and finds that most of the variation the judge reports across strata is present in the reference briefs themselves.

### *E. Verification*

Every table in this paper is recomputed from the released artifacts by a verification script distributed with the code, which re-derives each reported value from the corpus, the evaluation outputs, and the judge ratings, and reports any disagreement with the published figure. All tables reported as measured pass; the two exceptions — figures inherited from the v1 release and quantities requiring artifacts that were not retained — are marked as such in the text.

## IX. RESULTS

### *A. Revision trajectory*

Table VI reports every revision, scored by the protocol of Section VIII. **Cross-revision comparison is generally not like-for-like.** Each revision regenerated the corpus and therefore the test split; we confirmed by checksum that all eight formatted test splits are distinct files. Equal reference ceilings across revisions (v5 and v8 both report 0.701) are coincidental and do *not* indicate a shared split. The one exception is `sft_only_greedy`, `sft_v2`, and `sft_v2_sampled`, which were scored in a single evaluation run against one test split and are directly comparable to each other. Reward and accuracy deltas between revisions are consequently not valid measures of model change, and we do not report them as such. Within-revision comparisons against that revision's own ceiling are valid.

Three trends survive the caveat because they are large and monotone. Risk accuracy rises from 0.792 to 0.960 across the sequence. Unparseable headers fall to zero. Severe errors fall to 3, from a v4 excursion of 22. Reward, by contrast, does not improve — v1 records the highest reward in the series and v8 nearly the lowest. Section XI explains why these move in opposite directions.

*The zero-shot floor:* The base model, evaluated on the v8 split under the identical protocol, supplies a contemporaneous floor. It scores 0.000 exact risk accuracy — and, revealingly, 0.000 within-one-class accuracy with zero severe errors. Those three figures can co-occur only if every completion was unparseable: the base model emits no `RISK LEVEL`, `YOUR STATUS`, or `OVERALL RISK CLASSIFICATION` header at all, so all 500 predictions fall into the *unknown* bucket rather than being wrong. Overreaching accuracy is 0.006, three completions in 500.

This reframes the headline number. The gap between 0.000 and 0.960 is not the distance between poor and good clinical judgment; it is overwhelmingly the distance between unstructured prose and the corpus's brief format. Fine-tuning unambiguously taught the format, and the format is a precondition for the metric to register anything at all. What the metric cannot tell us — and what Section X addresses by other means — is how much clinical quality accompanies it.

The floor is more informative still for the composite reward, where the base model scores 0.422 against a ground-truth ceiling of 0.701. A model that never produces a readable risk header nonetheless earns 60% of what the reference briefs earn. Section XI takes this up. Normalized against floor and ceiling rather than against 1.0, `sft_v8`'s reward of 0.668 represents 88% of the achievable range.

Judge clinical accuracy peaks at v2 (4.38) and does not recover; v8 sits at 4.10. Actionability and clarity are flat from v2 onward. Given differing test splits we do not claim a regression, but the corpus revisions from v3 onward clearly did not improve judged clinical quality.

### *B. v8 in detail*

Per-register accuracy is uniform on both metrics: risk at 0.958, 0.964 and 0.958 for athlete, coach and sports-scientist respectively, and overreaching at 0.521, 0.503 and 0.488.

That uniformity is more informative than it appears, and it means different things for the two metrics. For risk level it is a positive control on the extractor. The three registers announce risk under different headers (`YOUR STATUS`, `RISK LEVEL`, `OVERALL RISK CLASSIFICATION`), so equal accuracy across them confirms the register-aware matcher is not silently failing on one surface form — a failure that would otherwise present as a genuine register-specific capability gap. It also shows the model holds risk assignment invariant under vocabulary substitution: the athlete register's lay rewriting does not change which class it assigns, only how it is worded.

For overreaching the same uniformity carries the opposite implication. Three registers with substantially different vocabularies, one of which replaces the clinical terms the metric

TABLE VI

ALL MODEL REVISIONS. EACH ROW IS EVALUATED AGAINST ITS OWN CONTEMPORANEOUS TEST SPLIT; THE VARYING CEILING CONFIRMS THE SPLITS DIFFER, SO COLUMNS ARE *not* COMPARABLE ACROSS ROWS. JUDGE SCORES ARE MEANS OVER 438 STRATIFIED RATINGS ON A 1–7 SCALE.

| Model | Ceiling | Reward | Risk | Within-1 | Severe | Unk. | Overr. | Words | Clin. | Clar. |
|---|---|---|---|---|---|---|---|---|---|---|
| *zero-shot floor*[‡] | 0.701 | $0.422 \pm 0.116$ | 0.000 | 0.000 | 0 | 500 | 0.006 | 250.0 | — | — |
| sft_only_greedy (v1) | 0.691 | $0.755 \pm 0.092$ | 0.792 | 0.974 | 10 | 3 | **0.758** | 153.1 | 3.00 | 5.75 |
| sft_only (v1, sampled) | 0.691 | — | — | — | — | — | — | — | 3.24 | 5.78 |
| sft_v2 | 0.691 | $0.675 \pm 0.104$ | 0.794 | 0.992 | 4 | 0 | 0.458 | 168.4 | **4.38** | **6.37** |
| sft_v2_sampled | 0.691 | $0.660 \pm 0.129$ | 0.702 | 0.948 | 13 | 13 | 0.352 | 165.7 | — | — |
| sft_v3 | 0.712 | $0.735 \pm 0.104$ | 0.842 | 0.994 | 3 | 0 | 0.518 | 176.0 | 4.09 | 6.36 |
| sft_v4 | 0.716 | $0.649 \pm 0.136$ | 0.856 | 0.956 | 22 | 0 | 0.482 | 178.8 | 4.23 | 6.28 |
| sft_v5 | 0.701 | $0.720 \pm 0.124$ | 0.934 | 0.982 | 8 | 1 | 0.552 | 176.0 | 4.23 | 6.41 |
| sft_v6 | 0.697 | $0.701 \pm 0.136$ | 0.942 | 0.972 | 3 | 11 | 0.512 | 171.2 | 4.11 | 6.29 |
| sft_v7 | 0.704 | $0.720 \pm 0.117$ | 0.948 | 0.986 | 3 | 4 | 0.554 | 173.5 | 3.89 | 6.32 |
| **sft_v8** | 0.701 | $0.668 \pm 0.099$ | **0.960** | **0.994** | 3 | **0** | 0.504 | 173.2 | 4.10 | 6.31 |

[‡]Zero-shot severe errors are 0 because all 500 completions were unparseable, not because predictions were accurate; the base model is evaluated on the v8 split and is therefore directly comparable to `sft_v8` only.

matches on with lay paraphrases, produce accuracies within 0.033 of one another and within 0.05 of chance on a four-class problem. A capability that transferred across registers would be expected to vary with them; a measurement artifact would not. Section XII establishes which of these it is.

### C. Stratified judge results

Table VII gives v8 judge scores by scenario type. Clinical accuracy spans 2.00 to 5.72 while clarity spans only 5.67 to 6.79. Weak strata therefore produce briefs that are well-formed and readable but clinically inconsistent: surface fluency is not a usable proxy for clinical correctness in this model's outputs.

The weak strata are not merely the conflicting-signal scenarios; they are the scenarios carrying documented contradictions. Comparing Table VII against Figure 3, the nine scenarios containing contradictory records average 3.31 clinical while the ten clean scenarios average 4.70, and the per-scenario contradiction rate correlates with judged clinical accuracy at Spearman $\rho = -0.67$ (95% bootstrap CI $[-0.87, -0.33]$, $p = 0.001$, $n = 19$). This is the signature predicted by Section IV, now measured rather than inferred and plotted in Figure 5: a constant label is correct whenever the sampled signals happen to agree with it and incoherent whenever they do not, and the judge penalizes exactly the strata where they do not.

Figure 5 shows that the relationship is closer to presence-versus-absence than to dose-response. The two groups separate almost completely — scenarios carrying contradictions span 2.00–4.26 while clean ones span 3.27–5.72, overlapping only in a narrow band — but *within* the affected group the contradiction rate predicts much less ($\rho = -0.55$, $n = 9$). `acwr_spike` carries contradictions in 85.6% of its records and scores 3.47, above `high_acwr_stable_physiology` at 12.2% and 2.00. A scenario appears to pay most of the cost as soon as it carries contradictory records at all; carrying more of them adds comparatively little.

The clean group spans 2.5 points on its own, so contradictions account for part of the stratified variation and not

TABLE VII

V8 JUDGE SCORES BY SCENARIO TYPE, 438 RATINGS, 1–7 SCALE, SORTED WEAKEST-CLINICAL-FIRST. STRATA MARKED † HAVE $n < 15$ AND ARE INDICATIVE ONLY.

| Scenario | Clinical | Action. | Clarity | $n$ |
|---|---|---|---|---|
| high_acwr_stable_physiol. | 2.00 | 3.78 | 5.67 | 9[†] |
| preseason_intensification | 2.67 | 4.76 | 6.05 | 21 |
| undertraining | 2.90 | 3.59 | 5.77 | 39 |
| monotony_problem | 3.25 | 4.88 | 6.04 | 24 |
| travel_jet_lag | 3.27 | 3.47 | 5.93 | 15[†] |
| double_session_accum. | 3.33 | 4.20 | 5.87 | 15[†] |
| acwr_spike | 3.47 | 5.83 | 6.72 | 36 |
| normal_progressive | 3.71 | 3.38 | 5.71 | 21 |
| heat_acclimatization | 3.75 | 5.33 | 6.33 | 24 |
| post_competition | 3.78 | 4.00 | 6.06 | 18 |
| fixture_congestion | 3.79 | 5.12 | 6.30 | 33 |
| taper | 4.26 | 4.30 | 6.22 | 27 |
| recreational_minimal | 4.67 | 4.33 | 6.33 | 3[†] |
| illness_return | 4.71 | 5.21 | 6.46 | 24 |
| youth_growth_spurt | 5.24 | 5.52 | 6.71 | 21 |
| early_overreaching | 5.29 | 5.24 | 6.57 | 21 |
| altitude_camp | 5.41 | 5.19 | 6.67 | 27 |
| overtraining syndrome | 5.60 | 6.36 | 6.79 | 42 |
| wellness_crash_normal_load | 5.72 | 5.72 | 6.78 | 18 |
| Overall | 4.10 | 4.90 | 6.31 | 438 |

all of it. `travel_jet_lag` and `double_session_accumulation` are clean by both detectors and still score near the bottom of the table. We read this as evidence that the two families of Section V are two mechanisms among several rather than the only one, and we have not characterized the others. What the figure does establish is directional and one-sided: carrying contradictions is sufficient to depress the stratum, and no scenario carrying them scores well.

*A reversal worth reporting:* The v1 release identified `overtraining_syndrome`, `heat_acclimatization`, and `youth_growth_spurt` as the weakest strata (2.2–2.6) and attributed this to override scenarios in which label intent diverges from sampled signals. In v8 these strata are

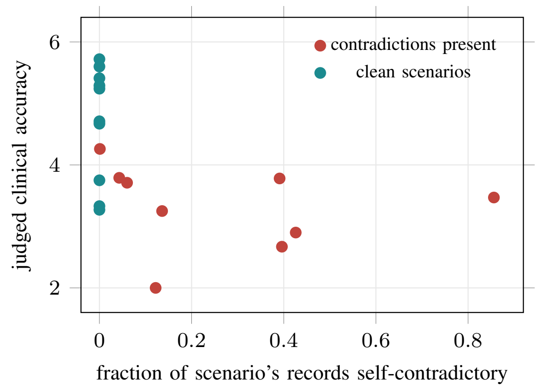


Fig. 5. Per-scenario contradiction rate against judged clinical accuracy, 19 scenarios ($\rho = -0.67$, 95% CI $[-0.87, -0.33]$, $p = 0.001$). Scenarios carrying contradictory records average 3.31; clean scenarios average 4.70. The same relationship holds against the ground-truth briefs (Section X), locating the cause in the corpus.

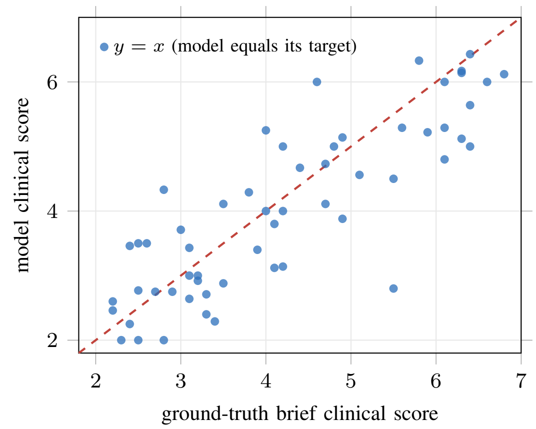


Fig. 6. Model clinical score against the clinical score of the ground-truth briefs it was trained to reproduce, for all 57 scenario-register cells (Table VIII). Points cluster along the identity line ($\rho = +0.83$, 95% CI $[+0.71, +0.89]$, mean offset $-0.19$): where the corpus is clinically weak the model is weak, and where the corpus is sound the model is sound. The stratified weakness of Table VII is inherited, not introduced.

among the strongest: OTS 5.60, youth growth spurt 5.24, heat acclimatization 3.75. The v1 diagnosis was correct about the mechanism and wrong about which scenarios it would persist in; corpus revisions repaired those three while the same mechanism continued to operate elsewhere. We report the reversal because the v1 characterization is public and should not be relied upon.

## X. Judge Calibration: The Model Reproduces Its Data's Clinical Quality

A stratified judge result of the kind in Table VII admits two readings. Either the model produces clinically weak briefs in those strata, or the judge scores those strata harshly regardless of what it is shown. These predict the same observation and cannot be separated by examining model outputs alone.

We separate them by scoring the corpus's own ground-truth briefs with the identical judge, prompt, and call path used in the main evaluation. If the reference briefs — the exact targets the model is trained to reproduce — score as low as the model's, the residual is not model error. We rated ten reference briefs for every scenario in every register: 57 cells, 570 ratings. Nineteen coach-register cells were additionally rated by a judge from a different vendor family. Table VIII reports every cell.

Four results follow.

**There is no judge ceiling.** Reference briefs reach 6.80 and multiple cells exceed 6.0. The judge assigns high scores when the briefs warrant them, so low scores elsewhere are not an artifact of judge severity.

**The model reproduces its data's clinical quality closely.** Across 57 cells the model's score tracks the ground-truth score at Spearman $\rho = +0.83$ (95% bootstrap CI $[+0.71, +0.89]$, $p < 10^{-4}$, $n = 57$), with a mean gap of $-0.19$ on a seven-point scale; the model equals or exceeds its reference briefs in 22 of 57 cells. Figure 6 plots every cell. The clinical weakness reported in Table VII is therefore overwhelmingly a property of the corpus rather than of the fine-tune. Where the reference briefs are clinically incoherent, the model faithfully reproduces that incoherence; where they are sound, it scores near them.

We do not claim the model has no failures of its own — the residual $-0.19$ is real, and individual cells such as `double_session_accumulation` at coach register show gaps of $-2.70$ — but the stratum-level pattern that Table VII reports is inherited, not introduced.

**Contradiction rate predicts reference-brief quality as strongly as it predicts model quality.** Restricting to the coach register, the nine scenarios carrying contradictions (Figure 3) average 3.34 ground-truth clinical against 5.35 for the ten clean scenarios, at Spearman $\rho = -0.69$ (95% CI $[-0.93, -0.31]$, $p = 0.001$, $n = 19$) — statistically indistinguishable from the coefficient we report between contradiction rate and *model* clinical accuracy in Section IX. This closes the causal chain: constant labels produce self-contradictory reference briefs, the reference briefs score poorly, and the model trained on them scores poorly in the same strata.

**The athlete register degrades clinical content in the reference briefs themselves.** Ground-truth clinical accuracy averages 3.52 for the athlete register against 4.40 for coach and 4.78 for sports scientist, and the athlete register is the lowest of the three in 16 of 19 scenarios. This is not a model effect — it is measured on the corpus's own output. The cause is visible in the generator: the audience adapter substitutes lay vocabulary for clinical terms, replacing “non-functional overreaching” with “overtraining warning” and “HRV” with “recovery score.” Section XI independently shows that this same substitution forfeits 0.172 of the 0.20 available terminology reward for every athlete brief. Two unrelated measures — a programmatic reward and a language-model judge — agree that the register adaptation removes clinical content rather than merely rephrasing it. For a system whose stated purpose includes communicating with athletes, this is a substantive finding and not a formatting detail.

TABLE VIII

JUDGE CALIBRATION ON GROUND-TRUTH BRIEFS, ALL 19 SCENARIOS × 3 REGISTERS, $n = 10$ PER CELL. GT IS THE REFERENCE-BRIEF CLINICAL SCORE; M IS THE V8 MODEL'S SCORE FOR THE SAME CELL. ACROSS ALL 57 CELLS THE MODEL TRACKS ITS TRAINING TARGETS AT SPEARMAN $\rho = +0.828$ WITH A MEAN GAP OF $-0.19$ ON THE 1–7 SCALE.

| | athlete | | coach | | sports scientist | | |
|---|---|---|---|---|---|---|---|
| Scenario | GT | M | GT | M | GT | M | Contra. |
| acwr_spike | 4.00 | 5.25 | 2.40 | 2.25 | 3.20 | 2.92 | 85.6% |
| undertraining | 2.20 | 2.46 | 2.50 | 2.77 | 2.40 | 3.46 | 42.6% |
| preseason_intensification | 3.20 | 3.00 | 3.30 | 2.71 | 3.40 | 2.29 | 39.6% |
| post_competition | 2.50 | 3.50 | 2.60 | 3.50 | 2.80 | 4.33 | 39.1% |
| monotony_problem | 2.70 | 2.75 | 4.10 | 3.12 | 4.90 | 3.88 | 13.6% |
| high_acwr_stable_physiol. | 2.30 | 2.00 | 2.50 | 2.00 | 2.80 | 2.00 | 12.2% |
| normal_progressive | 3.00 | 3.71 | 3.80 | 4.29 | 4.20 | 3.14 | 6.0% |
| fixture_congestion | 3.10 | 2.64 | 4.20 | 4.00 | 4.70 | 4.73 | 4.3% |
| taper | 3.50 | 4.11 | 4.70 | 4.11 | 5.10 | 4.56 | 0.1% |
| altitude_camp | 4.40 | 4.67 | 5.90 | 5.22 | 5.80 | 6.33 | 0 |
| double_session_accum. | 3.30 | 2.40 | 5.50 | 2.80 | 6.10 | 4.80 | 0 |
| early_overreaching | 3.10 | 3.43 | 6.40 | 6.43 | 6.10 | 6.00 | 0 |
| heat_acclimatization | 2.90 | 2.75 | 4.00 | 4.00 | 5.50 | 4.50 | 0 |
| illness_return | 3.50 | 2.88 | 6.30 | 5.12 | 6.80 | 6.12 | 0 |
| overtraining syndrome | 6.40 | 5.00 | 6.30 | 6.14 | 6.40 | 5.64 | 0 |
| recreational_minimal_data | 3.10 | 3.00 | 4.20 | 5.00 | 4.60 | 6.00 | 0 |
| travel_jet_lag | 3.90 | 3.40 | 2.20 | 2.60 | 4.10 | 3.80 | 0 |
| wellness_crash_normal_load | 4.80 | 5.00 | 6.60 | 6.00 | 6.30 | 6.17 | 0 |
| youth_growth_spurt | 4.90 | 5.14 | 6.10 | 5.29 | 5.60 | 5.29 | 0 |
| **Register mean** | **3.52** | **3.53** | **4.40** | **4.07** | **4.78** | **4.52** | |

### A. Cross-judge agreement

Nineteen coach-register cells were rated by both judges. Rank agreement on clinical accuracy is strong ($\rho = +0.76$, $p = 10^{-4}$, $n = 19$), but the level offset is substantial: the second judge scores 1.19 points lower on average, with a mean absolute difference of 1.29. The two judges therefore agree on *which* strata are weak while disagreeing on how weak.

This matters for how our numbers should be read. Every absolute judge value in this paper is specific to the judge that produced it; only the ordering is robust. Our substantive claims — that weak strata coincide with contradiction strata, that the model tracks its reference briefs, that the athlete register scores lowest — are all ordering claims and survive the change of judge. Any claim resting on an absolute threshold would not.

Agreement also varies by dimension: clinical accuracy $\rho = +0.762$, actionability $+0.695$, clarity $+0.540$. Clarity is the least reproducible of the three across judge families, which bears directly on the next point.

### B. Clarity is judge-dependent

Section IX observes that clinical scores span 2.00–5.72 while clarity stays nearly flat at 5.67–6.79, and reads this as evidence that surface fluency is decoupled from clinical correctness. Clarity is the dimension on which our two judges agree least ($\rho = +0.540$, against $+0.762$ for clinical accuracy), and the stricter judge penalizes internally contradictory briefs on clarity as well as on clinical accuracy — treating self-contradiction as an intelligibility failure rather than only a clinical one.

We therefore restrict the decoupling claim to the judge that produced our reported numbers, and offer it as a caution for single-judge stratified protocols generally: which evaluation dimensions appear independent of one another is partly a property of the judge, not only of the text.

**Scope.** This calibration covers ten briefs per cell with percentile bootstrap intervals, all 19 scenarios and all three registers for one judge, and 19 coach-register cells for a second. It rules out a judge ceiling, establishes the model-versus-data split corpus-wide, and quantifies cross-judge agreement. It does not include human expert ratings, which remain the strongest available addition to this evaluation.

## XI. THE REWARD FUNCTION IS MIS-SPECIFIED

### A. Definition

The composite reward assigns: 0.30 for a structured risk declaration matching `risk level: <class>` or `your status: <class>`, or 0.15 for a bare class word; up to 0.25 for recommendation specificity on a graded scale (percentage range 0.25, RPE threshold 0.22, time period 0.18, bare percentage 0.15, action verb 0.10); 0.20 for clinical overreaching terminology; 0.15 for an escalation trigger; and 0.10 for length in 200–450 words, or 0.05 for 100–600. The same function serves as the GRPO reward and as the primary evaluation metric.

### B. Decomposition of the ceiling

We scored the ground-truth briefs component-wise across all 40,000 records in three registers (Figure 7). The pooled mean reproduces the reported reference ceiling of 0.701 exactly.

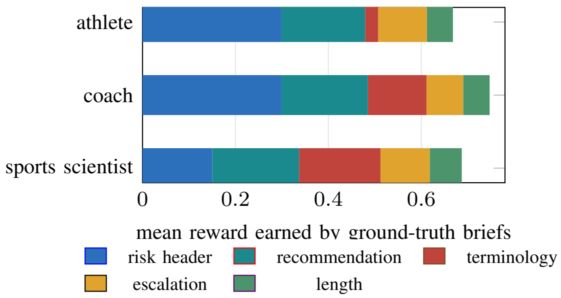


Fig. 7. Composite reward earned by the corpus's *own ground-truth briefs*, decomposed by component ($n = 40{,}000$ per register). Maximum available is 1.00; no register exceeds 0.75. The sports-scientist register forfeits half the risk component because its `OVERALL RISK CLASSIFICATION` header is unmatched; the athlete register forfeits 86% of the terminology component because its lay vocabulary removes the clinical strings; every register forfeits most of the length component because briefs fall below the 200-word band. The pooled mean, 0.701, is the reference ceiling reported throughout this paper.

Three components fail systematically, for three distinct reasons, all of them mismatches between the reward and the corpus's own conventions:

**Register header.** Sports-scientist briefs open `OVERALL RISK CLASSIFICATION:` in all 40,000 records. The reward matches only `risk level:` and `your status:`, so every sports-scientist brief forfeits 0.15 — one-third of the register's achievable margin. The *accuracy* extractor in the same evaluation script handles this header correctly. The reward and the accuracy metric therefore disagree about what a valid brief looks like, within one file.

**Register vocabulary.** The athlete register substitutes lay vocabulary for clinical terms by design (Section III-D) ("overtraining warning" for `non_functional_overreaching`). The reward requires the clinical strings, so athlete briefs earn 0.028 of an available 0.20 — a 0.172 systematic loss on a third of the corpus, incurred precisely by the register adaptation the system is built to perform.

**Length band.** The reward's full-credit band is 200–450 words. Register length targets are 150–250, 250–400, and 350–600 words, and realized means are 156, 177, and 196. Nearly every ground-truth brief lands in the partial band, earning 0.05–0.07 of an available 0.10.

### C. Consequences

Reward and risk accuracy move in opposite directions across revisions because they measure different things. Revisions improved the model's ability to emit the correct scenario constant, which accuracy sees and the reward largely does not; the reward is dominated by formatting properties the ground truth itself cannot satisfy.

This also supplies the mechanism for the v1 GRPO saturation result. A policy optimizing a reward whose ceiling is set by unearnable components has little exploitable headroom, and the components that remain exploitable are formatting artifacts rather than clinical quality. The original report concluded that SFT extracts all learnable signal when data and reward derive from the same rules. The stronger statement supported here is that the reward was substantially measuring register-formatting compliance, and was mis-specified against the very register conventions the corpus defines.

The zero-shot floor sharpens this considerably. The untuned base model, which emits no parseable risk header in any of 500 completions and classifies overreaching correctly three times, scores 0.422 against the ground-truth ceiling of 0.701. A metric on which unusable output earns 60% of what correct output earns has a usable dynamic range of 0.279, not 1.0, and the great majority of its scale is occupied by properties — some class word appearing somewhere, an action verb, a plausible length — that any fluent English continuation satisfies. The 0.246 that separates `sft_v8` from the base model is 88% of the range the metric can actually express, which is a far more useful summary than 0.668 against a notional 1.0.

Reporting a reference ceiling was the right precaution and remains insufficient. A ceiling below 1.0 tells you the metric is not normalized; it does not tell you *why*, and here the why is a defect rather than a property of the task. We recommend two checks before a composite reward is used as either a training signal or a headline metric: component-wise decomposition against ground-truth references, which reveals components the references cannot earn, and scoring an untrained or unrelated model to establish the floor, which reveals how much of the scale is occupied by properties the task does not require.

## XII. Overreaching Classification: A Metric That Cannot Read Correct Answers

v8 classifies overreaching state at 0.504 accuracy, uniformly across registers (0.521, 0.503, 0.488), and below the 0.575 rate obtainable by always predicting the majority class. Every revision from v2 onward sits below that baseline, while the earliest model reached 0.758. We initially reported this as a negative result and attributed it to a many-to-one scenario-to-class mapping. That account is wrong, and the correct one is a caution about metric design.

### A. Self-extraction: scoring a corpus against itself

The overreaching metric locates an ALL-CAPS classification section in a brief and string-matches clinical vocabulary within it. Its measurement is therefore contingent on how the corpus phrases that section. To test whether the metric can read correct answers at all, we passed each corpus's own ground-truth briefs through the unmodified extractor and scored them against their own labels. A ground-truth brief is by construction the correct answer, so a sound extractor returns 1.00. Whatever it returns instead is a ceiling on any model evaluated against that corpus: it is the accuracy a model would be reported as achieving if it reproduced its training data perfectly. Table IX reports the result for every corpus revision.

### B. The cause is a rationale layer, not a label mapping

The v1 corpus states its classification explicitly:

> **CLINICAL CLASSIFICATION:** Functional overreaching — short recovery period (3–7 days) will restore performance.

TABLE IX

SELF-EXTRACTION ACCURACY ON GROUND-TRUTH BRIEFS. THE v1 CORPUS IS FULLY READABLE; EVERY SUBSEQUENT CORPUS CAPS ANY MODEL AT APPROXIMATELY 0.55. MODEL SCORES ARE THOSE OF TABLE VI.

| Corpus | Self-extract | Unreadable | Model | % of ceiling |
|---|---|---|---|---|
| v1 | **1.000** | 0.0% | 0.758† | — |
| v2 | 0.550 | 45.0% | 0.458 | 83% |
| v3 | 0.550 | 45.0% | 0.518 | 94% |
| v4 | 0.551 | 44.9% | 0.482 | 87% |
| v5 | 0.552 | 44.8% | 0.552 | **100%** |
| v6 | 0.553 | 44.7% | 0.512 | 93% |
| v7 | 0.552 | 44.8% | 0.554 | **100%** |
| v8 | 0.551 | 44.9% | 0.504 | 91% |

†Trained on v1, evaluated on the v2 test split; see text.

From v2 onward, conflicting-signal classes carry a rationale paragraph in place of a classification:

> **CLINICAL CLASSIFICATION:** Load has normalised but overtraining markers persist. The low ACWR reflects accumulated chronic fatigue rather than detraining, and does not indicate recovery. *[label:* `overtraining_syndrome`*]*

The section is clinically informative and arguably better writing. It simply never names the class. The extractor, correctly anchored to the right section, finds no vocabulary to match and returns *unknown*. The effect is categorical for the classes that receive rationale treatment: `overtraining_syndrome` extracts correctly 0 times out of approximately 5,100 in every corpus from v2 to v8, in all three registers, without a single exception. `non_functional_overreaching` extracts at 0.33 and `normal_adaptation` at 0.49.

This is the same `conflict_rationale` machinery that renders the consistency checker of Section V blind to the contradiction records. A single component, added to explain declared signal conflicts, simultaneously disabled one deterministic guard and one evaluation metric. Both failures are silent.

### C. Consequences for the reported numbers

**The apparent collapse is a measurement change, not a capability loss.** Reported accuracy fell from 0.758 to 0.458 between the first two models because the readable fraction of the corpus fell from 1.00 to 0.55. Normalized against their own ceilings, the models sit at 83–100% and reach ceiling exactly at v5 and v7. The underlying trend is flat to improving, opposite in sign to the reported one.

**The first model's 0.758 is consistent with this account, and the completions confirm it.** That checkpoint was trained on v1, which teaches explicit class statements, and evaluated on the v2 test split; it therefore emits extractable classifications while being scored against v2 labels, and can exceed what a v2-faithful model could achieve. Two independent lines of evidence support the attribution. Its checkpoint predates the v2 corpus by two months, so v1 is the only corpus it could have been trained on. More directly, applying the extractor to the stored completions themselves (Figure 8) shows it names a

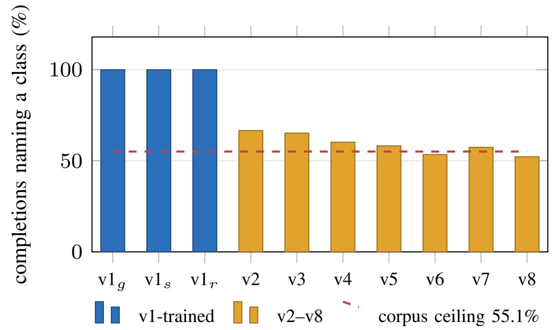


Fig. 8. Fraction of stored completions in which the overreaching extractor finds a named class. $v1_g$, $v1_s$ and $v1_r$ are the greedy, sampled and GRPO checkpoints trained on v1; all three are fully readable. Every v2-onward model declines toward the readable fraction of its own training corpus (dashed line). Later models learned the rationale phrasing more thoroughly, so their outputs became progressively less legible to the metric: reported accuracy fell as fidelity to the training data rose.

class in 100% of its 500 outputs, as do both surviving v1-era reinforcement-learning artifacts, while every v2-onward model names one in only 52–67%.

Figure 8 also shows a trend the reported accuracies conceal: the named-class rate declines almost monotonically from 66.6% at v2 to 52.2% at v8, converging on the corpus's own 55% readable fraction. Later models learned the rationale style more thoroughly, so their outputs became progressively less legible to the metric. Reported overreaching accuracy fell as fidelity to the training data rose.

**The majority-class comparison does not survive.** A constant predictor scores 0.575 because it emits a literal class phrase on every example. A model faithful to the corpus caps at 0.551. Scoring below the majority-class baseline is therefore not evidence of failure here: the baseline is unreachable by any model that learned its training data. We withdraw that comparison.

**A reward component is affected identically.** Section XI reports that ground-truth coach briefs earn only 0.126 of the 0.20 available overreaching-terminology reward. The cause is the same: briefs whose classification section contains rationale prose do not contain the clinical vocabulary the reward function matches. One generator change degraded a classification metric, a reward component, and a consistency guard, through a single mechanism.

### D. What we can and cannot conclude

We can conclude that reported overreaching accuracy in this project measures extractability at least as much as classification, that cross-revision comparisons of it are invalid because the ceiling moved, and that the released model performs at 91% of what its corpus permits under this metric.

We cannot conclude that the model classifies overreaching well. The metric that would establish it does not work on this corpus, and we have not replaced it. A label-based evaluation — scoring the model's classification against ground truth without routing through free-text extraction — would settle the question and is the obvious next step. Until then, downstream users should not rely on the overreaching field, and the honest statement is not that the model fails at this task but that we cannot currently measure whether it succeeds.

**Generalization.** A metric that parses model output for a target string should be validated against reference outputs before use, exactly as a classifier is validated against a held-out set. Self-extraction is a one-line check: score the ground truth with the metric and confirm it returns the ground truth. Had we run it once at v2, we would have caught this immediately instead of reporting a capability loss across seven revisions.

## XIII. Limitations

**Cross-revision comparisons are invalid.** Every revision regenerated the corpus and test split. Table VI is a record of the development sequence, not a controlled ablation. No claim in this paper rests on a cross-revision delta.

**Thin strata.** Three strata in Table VII have $n < 15$, including both the weakest ($n = 9$) and one of the strongest ($n = 3$) cells. These are marked and should not be compared.

**No human validation.** All quality judgments come from language-model judges. Section X establishes rank agreement between two vendor families ($\rho = +0.762$ on clinical accuracy) and rules out a judge ceiling, but the two judges differ by 1.19 points in level, clarity is the least reproducible dimension across them, and no brief in this work has been rated by a practising sports scientist. Absolute judge values should be read as judge-specific; our substantive claims are ordering claims. Expert validation of even a small subset would be the strongest available addition to this evaluation and remains future work; we note that it would test something the present evaluation cannot, since two language-model judges agreeing with each other is weaker evidence than one domain expert.

**Rule-grounded, not outcome-validated.** Labels are correct with respect to published criteria, not validated against athlete injury outcomes.

**Fully synthetic.** No real athlete data was used at any stage. The corpus inherits every gap in the encoded criteria and does not reflect the messiness of real monitoring.

**Reachability audit is incomplete.** It covers numeric ACWR and monotony targets only. Scenarios whose labels rest on HRV-day counts or wellness composites are uncovered, including the defect of Section V.

**Reachability history is measured against current declarations.** Figure 4 evaluates every corpus revision against the scenario definitions as they stand today. It establishes that the released corpus satisfies its own declarations and that earlier corpora did not, but it cannot distinguish a repair made by correcting a sampler from one made by moving a declared target to meet the data. The scenario definitions were not under version control during development, so we cannot resolve this retrospectively and do not claim which mechanism produced each repair in Figure 4. The v8 row is exempt: the scenario definitions were last modified two days before the released corpus was generated and have not been touched since, so the released corpus is audited against the definitions that actually produced it. The caveat applies only to the historical rows. The audit's value as a build gate is unaffected — it compares declarations against output at the time it runs — but its use here as a historical record should be read with that limitation in mind. Placing generator definitions under version control is a straightforward precondition for the kind of retrospective audit this paper otherwise recommends.

**The label-agreement audit on v1 is approximate.** The rejection rate reported in Section V-D is computed by an audit mode that infers HRV-day counts and wellness composites from brief text rather than from the generator's metric values, so 4.1% is an estimate rather than the rate the in-generator check would have produced.

**Comparison models are not evaluated contemporaneously.** The v8 evaluation includes the untuned base model as a floor (Table VI), scored on the v8 split under the identical protocol. It does not include the prompted comparison models — Qwen 2.5 7B Instruct and Mistral 7B Instruct v0.3 — that the v1 release evaluated on a different corpus; those figures are not carried forward. Because the base model produces no parseable output at all, the floor establishes that fine-tuning taught the format but says little about how a well-prompted general model would fare. A few-shot prompted baseline would be a more demanding comparison and is not reported here.

**Forgetting is uncharacterized for the released model.** The v1 release measured general and clinical-reasoning benchmarks before and after fine-tuning and found substantial degradation: MedQA fell from 61.7 to 55.8, MMLU clinical knowledge from 77.4 to 66.0, MMLU college medicine from 63.6 to 59.5, and MMLU philosophy — the subject least related to the training domain, included as a control — from 71.7 to 59.8. That the largest drop occurred on the control subject indicated general capability loss rather than a targeted reshaping of clinical knowledge.

Those benchmarks were not re-run for v8, and we do not carry the v1 figures forward. They were measured on a model trained with a reinforcement-learning stage the released model does not have, on a corpus seven revisions removed from the released one. The direction of the effect is unlikely to have reversed — task specialization on 20,000 in-domain examples is the same procedure in both cases — but its magnitude for v8 is unknown. Users should treat the released model as a specialist of uncharacterized general capability and should not rely on it for clinical or general question answering outside its task.

## XIV. Recommendations

The checks below are each cheap, each would have caught a defect this project carried through multiple training cycles, and none is standard practice.

**1. Audit declared-target reachability.** For every generation parameter a scenario declares, compare the declared range against the distribution the sampler actually produces. This defect class is invisible to label-consistency validation by construction — the label is consistent with whatever was sampled; it is the *declaration* that is unreachable, and nothing compares the two. The check is a few lines, exits nonzero on failure, and can gate a build. Nine targets were defective in our v1 corpus and each was found separately and expensively over the following six revisions.

**2. State label provenance explicitly.** For each label, document whether it is computed from the features released alongside it or assigned independently of them. Where a label is a generator constant, classification accuracy measures recovery of the generator's configuration, not the capability the label names, and should be reported that way. Existing documentation frameworks [36]–[38] ask how data was collected, by whom, and for what purpose; for synthetic corpora we suggest they should also ask what each label is a function of.

**3. Validate extraction metrics against reference outputs.** Any metric that parses generated text for a target string should first be run on the corpus's own ground truth. If it cannot recover the ground-truth label from the ground-truth text, it is measuring phrasing rather than the property it names, and its self-extraction rate — not 1.0, and not the majority-class rate — is the correct denominator for model scores. This is a one-line check. Running it once would have saved us seven revisions of misreading a generator change as a capability loss.

**4. Decompose composite rewards against ground truth.** A reference ceiling below 1.0 is a warning, not a diagnosis. Score the reference outputs component-wise: a component the references cannot earn is a metric defect, not task difficulty. In our case three of five components were mis-specified against the corpus's own register conventions, which both capped the metric and supplied a mundane explanation for a previously published reinforcement-learning saturation result.

**5. Calibrate model-based judges against ground truth.** A judge score is interpretable as model quality only if the judge would score a correct answer highly, and nothing in a standard protocol establishes that. Scoring the corpus's own reference outputs with the identical judge separates model error from data error, supplies a per-stratum ceiling, and costs one additional judge run. It also surfaced, in our case, a register-level defect the main evaluation could not have found.

**6. Do not let a guard's scope be set by the data it inspects.** Where a consistency check exempts records on the basis of a field in those records, the check can be satisfied by editing the field rather than repairing the condition, and no artifact distinguishes the two. Keep exemption criteria in versioned configuration, and log exemptions rather than silently applying them.

**7. Pair deterministic and model-based evaluation, and treat neither as evidence about the other's blind spots.** A deterministic checker is exact within the defect taxonomy it encodes and silent outside it, in a way indistinguishable from a clean result. A model-based judge is noisier and not exhaustive, but is not blind by construction to classes nobody anticipated. In this project three deterministic guards passed every one of 6,433 self-contradictory records while the judges described the defect unprompted.

## XV. Conclusion

LoadBrief attains strong headline numbers — 0.960 exact risk accuracy, 0.994 within one class, 3 severe errors in 500, against a base model that scores zero — on a task for which no prior dataset existed. We have argued that the headline numbers measure recovery of a generator configuration rather than clinical inference, because both ground-truth labels are per-scenario constants; that this construction produces two disjoint defect families affecting 16.1% of records, every one of which passes all three of the corpus's own consistency guards; that the model's apparent clinical weakness is largely its faithful reproduction of those defective briefs, since across 57 scenario-register cells it tracks the ground-truth briefs at $\rho = 0.83$ and sits only 0.19 points below them; that the composite reward's ceiling is a mis-specification artifact rather than a measure of difficulty; and that a seven-revision decline in overreaching accuracy reflects a metric that stopped being able to read correct answers rather than a model that stopped producing them.

We release the corpus and model unmodified, with these findings documented, so the artifacts and the analysis remain consistent. The methodological claim is that rule-generated data is not self-validating. "Correct by construction" guarantees only that the label agrees with the rule that produced it. It says nothing about whether the rule's output agrees with the data rendered alongside it, whether the declared target was reachable, or whether the metric evaluating the result measures the intended property. Each of those required a separate check, and in this project each went unchecked through multiple training cycles.

## Ethics and Broader Impact

LoadBrief produces training-modification recommendations and medical-escalation triggers for athletes. It is a research artifact and is not a medical device.

No human subjects were involved and no real athlete data was used at any stage; the corpus is generated entirely by simulation from published thresholds. There are correspondingly no privacy or consent considerations in the data, and equally no evidence that the corpus reflects real monitoring distributions.

The principal risk is deployment. A model whose recommendations are correct with respect to encoded criteria may still be wrong about an athlete, and this paper documents specific mechanisms by which its outputs can be internally contradictory (Sections V–V-B) and clinically inconsistent in ways that surface fluency conceals (Section X). The underlying sports-science constructs are themselves contested [4], [5], and no output of this system has been validated against injury or illness outcomes or reviewed by a practising clinician. Acting on a brief that advises a 40–50% load reduction, or that does not escalate when escalation is warranted, carries real cost to an athlete.

We therefore release the model and corpus for research on structured generation and synthetic-data quality, and state plainly that they should not inform decisions about a real person's training or health. The dataset card and model card carry the same statement alongside the specific defects documented here, so that the limitations travel with the artifacts rather than only with this paper.

## Availability

Model: https://huggingface.co/tyhob/loadbrief. Dataset: https://huggingface.co/datasets/tyhob/loadbrief-50k. Code: https://github.com/tyhobbs/LoadBrief, including the reachability audit, the self-extraction check, the judge-calibration harness, and a verification script that recomputes every table in this paper from the released artifacts. Model weights derive from Llama 3 and are governed by the Meta Llama 3 license.

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